

Regular Expression

□ As discussed earlier that a^* generates

$\Lambda, a, aa, aaa, \dots$

and a^+ generates $a, aa, aaa, aaaa, \dots$, so the language $L_1 = \{\Lambda, a, aa, aaa, \dots\}$ and $L_2 = \{a, aa, aaa, aaaa, \dots\}$ can simply be expressed by a^* and a^+ , respectively.

a^* and a^+ are called the regular expressions (RE) for L_1 and L_2 respectively.

Note: a^+, aa^* and a^*a generate L_2 .

Recursive definition of Regular Expression(RE)

Step 1: Every letter of Σ including Λ is a regular expression.

Step 2: If r_1 and r_2 are regular expressions then

1. (r_1)

2. $r_1 r_2$

3. $r_1 + r_2$ and

4. r_1^*

are also regular expressions.

Step 3: Nothing else is a regular expression.

Defining Languages (continued)...

□ Method 3 (Regular Expressions)

- Consider the language $L = \{\Lambda, x, xx, xxx, \dots\}$ of strings, defined over $\Sigma = \{x\}$.

We can write this language as the Kleene star closure of alphabet Σ or $L = \Sigma^* = \{x\}^*$

this language can also be expressed by the regular expression x^* .

- Similarly the language $L = \{x, xx, xxx, \dots\}$, defined over $\Sigma = \{x\}$, can be expressed by the regular expression x^+ .



□ Now consider another language L , consisting of all possible strings, defined over $\Sigma = \{a, b\}$. This language can also be expressed by the regular expression $(a + b)^*$.

□ Now consider another language L , of strings having exactly double a , defined over $\Sigma = \{a, b\}$, then its regular expression may be

$$b^* a a b^*$$

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- Now consider another language L , of even length, defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$$((a+b)(a+b))^*$$

- Now consider another language L , of odd length, defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$$(a+b)((a+b)(a+b))^* \text{ or } ((a+b)(a+b))^*(a+b)$$

Remark



- It may be noted that a language may be expressed by more than one regular expressions, while given a regular expression there exist a unique language generated by that regular expression.

□ Example:

□ Consider the language, defined over $\Sigma = \{a, b\}$ of words having at least one a , may be expressed by a regular expression $(a+b)^*a(a+b)^*$.

□ Consider the language, defined over $\Sigma = \{a, b\}$ of words having at least one a and one b , may be expressed by a regular expression

$$(a+b)^*a(a+b)^*b(a+b)^* + (a+b)^*b(a+b)^*a(a+b)^*.$$

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- Consider the language, defined over $\Sigma = \{a, b\}$, of words starting with double a and ending in double b then its regular expression may be $aa(a+b)^*bb$
 - Consider the language, defined over $\Sigma = \{a, b\}$ of words starting with a and ending in b OR starting with b and ending in a, then its regular expression may be $a(a+b)^*b + b(a+b)^*a$

TASK

- Consider the language, defined over $\Sigma=\{a, b\}$ of **words beginning with a**, then its regular expression may be $a(a+b)^*$
- Consider the language, defined over $\Sigma=\{a, b\}$ of **words beginning and ending in same letter**, then its regular expression may be $(a+b)+a(a+b)^*a+b(a+b)^*b$

TASK

- Consider the language, defined over $\Sigma=\{a, b\}$ of **words ending in b**, then its regular expression may be $(a+b)^*b$.
- Consider the language, defined over $\Sigma=\{a, b\}$ of **words not ending in a**, then its regular expression may be $(a+b)^*b + \Lambda$. It is to be noted that this language may also be expressed by $((a+b)^*b)^*$.

SummingUP Lecture 3

RE, Recursive definition of RE, defining languages by RE, $\{x\}^*$, $\{x\}^+$, $\{a+b\}^*$, Language of strings having **exactly one aa**, Language of strings of **even length**, Language of strings of **odd length**, RE defines unique language (as Remark), Language of strings having **at least one a**, Language of strings having **at least one a and one b**, Language of strings **starting with aa and ending in bb**, Language of strings **starting with and ending in different letters**.